# Towards a Human-Guided AutoML Application for Democratizing Predictive Modeling in Clinical Research

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**Abstract**

Clinical research faces a significant gap between advanced machine learning capabilities and their practical adoption by non-technical researchers. In addition, existing AutoML tools treat clinical data as generic datasets, ignoring the methodological particularities of patient registries, observational studies and clinical trials. To address this, we present HAMELIN, an AutoML application under active development that bridges this gap through a protocol-aware interface aligned with human-guided machine learning principles. HAMELIN enables clinical researchers to build predictive models without coding, incorporating their domain knowledge to guide every single stage of the entire process. The system was designed through iterative collaboration with clinical research professionals using real clinical protocols. This paper describes the design principles, architecture and implementation of HAMELIN, and evaluates its alignment with human-guided machine learning criteria.

**Index Terms**

machine learning for healthcare, automated machine learning, human-guided machine learning, clinical research, protocol-aware systems

## I. Introduction

The rapid integration of artificial intelligence (AI) into medicine faces a key challenge: clinicians have domain knowledge but lack technical skills for complex machine learning (ML) pipelines, while data scientists can build advanced models but often lack the clinical context [1]. This expertise gap hinders the adoption of AI in clinical research.

Automated machine learning (AutoML) platforms have emerged as promising solutions by automating complex tasks such as feature engineering, algorithm selection, and hyperparameter optimization, enabling users without extensive programming expertise to build predictive models [2]. However, current tools fail to bridge this gap for clinical researchers [3]. They treat datasets as generic matrices, ignoring structures inherent to clinical research, such as patient registries, observational studies, and clinical trials. They are not protocol-aware, making it difficult to distinguish between variables that define study populations (inclusion criteria) and those measuring outcomes (endpoints). Moreover, their black-box nature complicates interpretation by researchers. Therefore, incorporating domain knowledge into the modeling process is essential [4].

To address these limitations, we present Human-guided Automated Machine Learning for Clinical Studies (HAMELIN), a no-code, protocol-aware AutoML platform grounded in human-guided machine learning (HGML) principles, which emphasize integrating human expertise throughout the ML pipeline. HAMELIN provides a collaborative environment where clinicians and data scientists work together using a shared language. By explicitly defining variables, distinguishing endpoints from exposures, and separating interventions from covariates, HAMELIN facilitates the development of predictive models that are both clinically coherent and aligned with the structural requirements of research designs. This approach effectively bridges the gap between technical expertise and clinical domain.

The development of HAMELIN followed a participatory design approach, involving clinical research professionals (CRPs) from the Clinical Trials Agency of IDIVAL (Marqués de Valdecilla University Hospital) throughout the design and refinement

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process. Four real clinical protocols served as use cases: COLCHI-COVID (a clinical trial evaluating colchicine in COVID-19 patients), EPA-Registro COVID-19 (a patient registry), UPPS (an observational study on pressure injuries in neurosurgery), and VACAB-2022 (a clinical trial on bacterial vaccines for prosthesis infections). These protocols grounded the interface design in authentic clinical research scenarios and ensured that the system's terminology, workflow, and validation mechanisms align with actual clinical practice.

The remainder of this paper is organized as follows. Section II provides background on clinical research methodologies, reviews related work in AutoML, human-centered artificial intelligence (HCAI), HGML, and introduces Ludwig as the underlying AutoML engine. Section III presents the participatory design process, the architecture, and the user interface HAMELIN, detailing each component of the tab-based workflow. Section IV evaluates the system's alignment with HGML criteria, identifying strengths and limitations. Section V discusses design decisions, trade-offs, and implications for future AutoML systems. Finally, Section VI concludes the paper and outlines directions for future work.

## II. Background

### A. Clinical Research Methodologies

Clinical research employs distinct frameworks or protocols for their research, each with specific design principles and analytical constraints [5]. Patient registries collect long-term data from people who share the same medical condition. These registries are typically observational and aim to understand disease history, treatment patterns, and long-term outcomes. Observational studies test specific hypotheses about exposure-outcome associations without intervention, requiring careful adjustment for confounding variables and selection bias. Clinical trials evaluate interventions through randomization, blinding and rigorous protocols defining inclusion/exclusion criteria, temporal phases (screening, baseline, intervention, follow-up, endpoint evaluation) and endpoint hierarchies (primary endpoints for efficacy, secondary endpoints for safety and exploratory outcomes).

These distinctions impose specific requirements on predictive modeling that generic ML pipelines ignore [6]. For instance, in a clinical trial, variables collected during screening (e.g., blood pressure to determine eligibility) have different meanings than the same variables collected at follow-up (e.g., blood pressure as an outcome). A model trained without awareness of these temporal and semantic constraints risks producing invalid results, such as data leakage, using information collected after the outcome to predict that outcome, or confounding, failing to account for variables that influence both exposure and outcome.

### B. Related Work

AutoML emerged as a need to bridge the gap between the growing demand for ML solutions and the limited availability of skilled data scientists in the mid-2010s [7]. By automating tasks such as feature engineering, algorithm selection, and hyperparameter optimization, AutoML platforms enable domain experts to develop predictive models without extensive programming expertise. This automation shifts the focus from technical implementation to strategic problem-solving, allowing users to concentrate on formulating research questions, interpreting results, and making informed decisions. The AutoML ecosystem has since expanded significantly, encompassing a diverse range of tools tailored to different user needs and organizational contexts. Commercial platforms such as Amazon SageMaker Autopilot [8], Azure Automated ML [9], and Google AutoML [10], among others, provide enterprise-grade solutions with intuitive interfaces, dedicated support, and seamless integration into existing workflows, making them attractive for organizations lacking in-house data science teams. In parallel, free and open-source frameworks such as Auto-sklearn [11], AutoGluon [12], H2O [13], and Ludwig [14], among others, have driven innovation through community collaboration, offering transparent, customizable, and freely accessible alternatives that support both academic research and practical applications. However, existing AutoML platforms remain largely domain-agnostic and fail to address the specific requirements of clinical research. They cannot distinguish variable hierarchy (primary endpoints vs. secondary endpoints vs. exposures), handle complex inclusion/exclusion logic that defines study populations, or recognize temporal data disposition across protocol phases. No tool to date adapts to the distinct methodological requirements of patient registries, observational studies and clinical trials within a unified interface.

Addressing this gap requires rethinking the relationship between automation and human expertise. HCAI [15] emphasizes designing AI systems that augment human capabilities rather than replace them, prioritizing human control, transparency, and trust. The HCAI framework (Figure 1) proposes a two-dimensional design space: the level of automation (from fully manual to fully autonomous) and the level of human control (from high human control to high computer control). Effective HCAI systems balance these dimensions, providing automation where appropriate while preserving meaningful human oversight. In clinical research, this balance is particularly critical: while automation can accelerate model development, clinicians must retain control over decisions that affect patient safety, study validity, and regulatory compliance. HCAI principles thus guide the design of HAMELIN by ensuring that automation serves CRPs rather than obscuring the modeling process.

Building on these HCAI principles, HGML [16] operationalizes human-centered design through specific interaction patterns where domain experts actively guide and influence the learning process. Rather than treating automation as a black box, HGML promotes collaborative workflows where experts validate intermediate decisions and shape model behaviour. The HGML

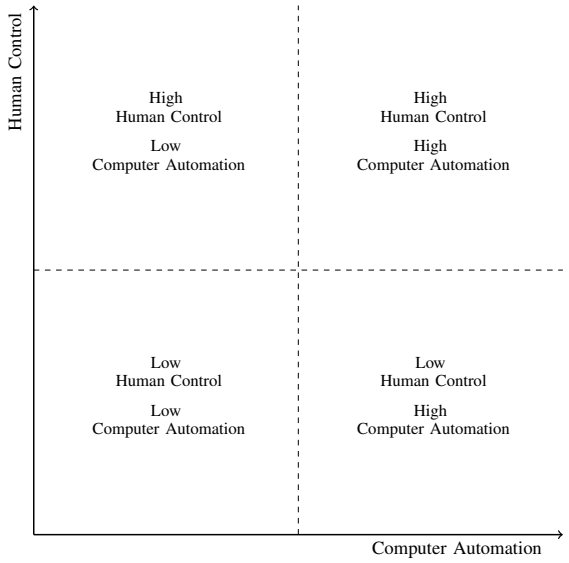



Fig. 1. Two-dimensional HCAI framework of human control and computer automation.

framework defines concrete tasks that domain experts should be able to perform across three categories: data use (variable selection, variable requirements, instance selection), model development (model selection, parameter settings, solution design), and model interpretation (model assessment, model comparison, parameter comparison). These tasks provide a structured framework for evaluating how well an AutoML system supports domain expert involvement. While HCAI provides high-level design principles for human-AI collaboration, HGML translates these principles into actionable criteria grounded in ML workflows, making it particularly suitable for assessing systems that aim to democratize predictive modeling in specialized domains.

### C. Ludwig AutoML

Among existing AutoML frameworks, Ludwig [14] presents an interesting starting point for building HGML systems. Ludwig automates the full ML workflow, including end-to-end model training, algorithm selection based on data characteristics, performance evaluation with multiple metrics and validation schemes, hyperparameter optimisation and inference on new data. Being YAML-based, a human-readable format, it allows pipelines to be specified declaratively without programming, making configurations transparent and editable. This declarative approach aligns well with HGML principles, as it enables domain experts to understand and potentially adjust the modeling pipeline. Therefore, the choice of Ludwig as a backend for HAMELIN is motivated by its ease of configuration, strong community support, and active open-source development, which ensure long-term maintainability [17].

Furthermore, Ludwig's explicit handling of data preprocessing decisions, particularly for missing data, provides the transparency and control required for HGML systems. Missing data is a pervasive challenge in clinical datasets, where values may be absent due to patient dropout, measurement errors, or protocol deviations. Ludwig addresses this by offering domain experts explicit control over how missing values are handled. For input features, users can choose to replace missing values with a constant, with the most frequent value or average value, propagate values forward or backward from adjacent time points, or remove incomplete cases entirely. For output features (the target variable), Ludwig's default behavior is to exclude cases with missing ground truth, as supervised learning requires known outcomes to train the model. This flexibility is particularly valuable in clinical research, where the appropriate treatment of missing values depends on the clinical context, study protocol, and the mechanism causing the missingness, and where domain experts must retain control over such critical decisions.

## III. Application

HAMELIN is a no-code, protocol-aware AutoML application under development specifically designed to operationalize HGML principles for clinical research. The system design was guided by mockups based on real clinical research protocols provided by CRPs. This ensures that the interface terminology, workflow structure, and validation mechanisms align with actual clinical research practices.

### A. Participatory Design Process

The development of HAMELIN followed a participatory design approach, involving CRPs throughout the design and refinement process.

Four real clinical protocols were served as use cases to ground the interface design in authentic clinical research scenarios: one patient registry, one observational study, and two clinical trials. These are the details of each protocol that were considered to ensure the interface was tailored as closely as possible to the required needs.

- **EPA-Registro COVID-19**: A multicenter patient registry documenting clinical characteristics, treatments, and outcomes of COVID-19 patients. The registry collected demographic variables (age, sex), comorbidities (hypertension, diabetes, cardiovascular disease), clinical variables (symptoms, laboratory values), treatments (antivirals, corticosteroids, oxygen therapy), and outcomes (hospital discharge, ICU admission, death).
- **UPPS (Pressure Injuries in Neurosurgery)**: An observational study investigating the incidence and risk factors of pressure injuries in neurosurgery patients undergoing procedures longer than 3 hours. The protocol specified variables related to patient characteristics (age, sex, BMI, Braden scale score), intraoperative factors (surgical duration, patient positioning, use of pressure-relief surfaces, intraoperative temperature), and postoperative outcomes (presence/absence of pressure injuries, injury severity).
- **COLCHI-COVID**: A randomized, double-blind, placebo-controlled clinical trial evaluating the efficacy and safety of colchicine in hospitalized COVID-19 patients. The protocol defined primary endpoints (clinical improvement at day 14), secondary endpoints (mortality, ICU admission, adverse events), and detailed inclusion/exclusion criteria (age $\geq$ 18 years, confirmed COVID-19, no contraindications to colchicine).
- **VACAB-2022**: A randomized clinical trial evaluating the efficacy of mucosal bacterial vaccines (autovaccines) compared to suppressive antibiotic therapy in patients with hip or knee prosthesis infections. The protocol defined primary endpoints (infection recurrence at 12 months), secondary endpoints (quality of life, adverse events), and inclusion/exclusion criteria (confirmed prosthesis infection, no immunosuppression, no allergy to vaccine components).

The design process was iterative and collaborative. Initial mockups were developed based on the analysis of these clinical protocols and periodically refined through feedback sessions with CRPs. These mockups served to validate the proposed terminology, workflow structure, and interaction patterns before full implementation (Figure 2).

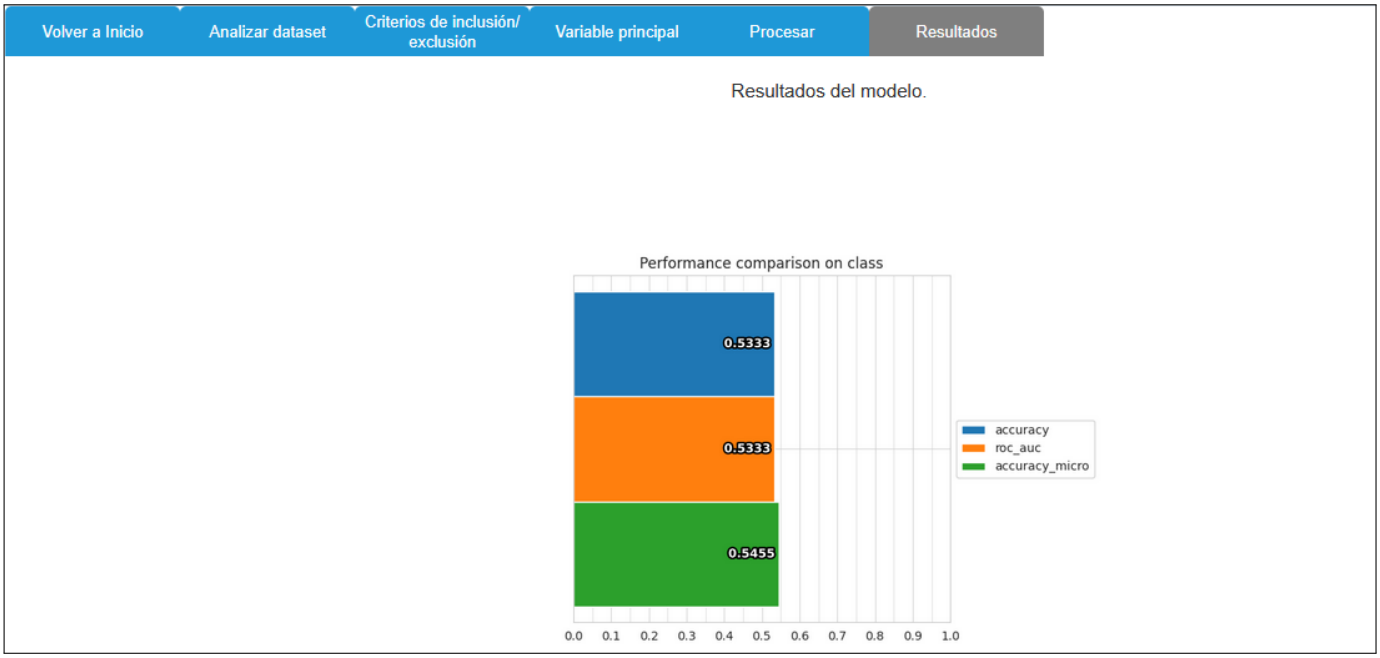



Fig. 2. Early mockup of the Results tab, illustrating its layout and how it relates to the other tabs in the interface (in Spanish).

### *B. System Architecture*

The system architecture follows a three-layer modular design (Figure 3). The interface layer provides a graphical environment for dataset loading, configuration, and visualization. The core logic layer connects the interface to the AutoML engine, handling validation, command translation, configuration file generation, workflow coordination, and error handling. The AutoML layer utilizes Ludwig (v0.10.4) to perform automated training, algorithm selection, hyperparameter optimization, and evaluation.

This modular architecture ensures that the interface logic is separated from the AutoML engine. In principle, Ludwig could be replaced with another AutoML framework without requiring changes to the interface, although some adaptation of the Core Logic Layer would be necessary.

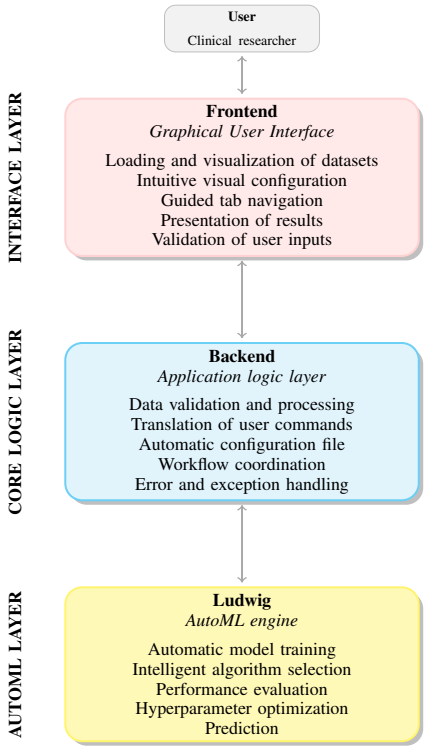



Fig. 3. Layered system architecture of HAMELIN

### C. User Interface and Workflow

The interface employs a tabbed navigation system that guides users through the entire ML pipeline. This design provides clear waypoints while maintaining the analytical context, with each tab representing a distinct phase that follows the logical progression typically found in clinical research protocols. A key feature of HGML is the ability to refine the model iteratively based on initial results. The HAMELIN interface supports this through a reconfiguration mechanism that allows users to revisit earlier steps and modify their choices based on emerging insights. After reviewing results, users can return to previous tabs to adjust the target variable, refine inclusion/exclusion criteria, select alternative missing data strategies, modify hyperparameter search spaces, or choose different evaluation metrics. Each modification triggers a model re-training cycle with updated results, enabling CRPs to explore the modeling space systematically, guided by clinical intuition and domain knowledge.

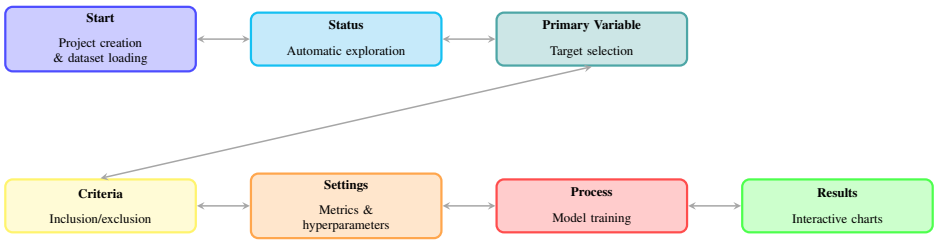



Fig. 4. End-to-end clinical machine learning workflow in HAMELIN.

The proposed workflow consists of seven sequential stages (Figure 4). The process initiates with project initialization and dataset loading, followed by automated exploratory data analysis to generate descriptive statistics, assess missing data patterns, and identify outcome variables. Subsequently, users specify the primary variable, with the system automatically determining the task type (classification or regression). Data filtering is performed through the definition of inclusion and exclusion criteria via a structured interface. The configuration stage encompasses the specification of evaluation metrics, missing data imputation strategies, and hyperparameter ranges. Model training is executed with progress tracking, and performance evaluation is presented through comprehensive visualizations including performance metrics, graphical representations, and confusion matrices for classification tasks (Figure 5). This iterative refinement process proves particularly valuable in clinical scenarios: for instance, if initial results reveal low sensitivity in detecting at-risk patients, a CRP might return to the Settings

tab to adjust the evaluation metric; alternatively, if feature importance analysis identifies a critical variable with substantial missing data, the imputation strategy can be reconsidered to improve model performance.

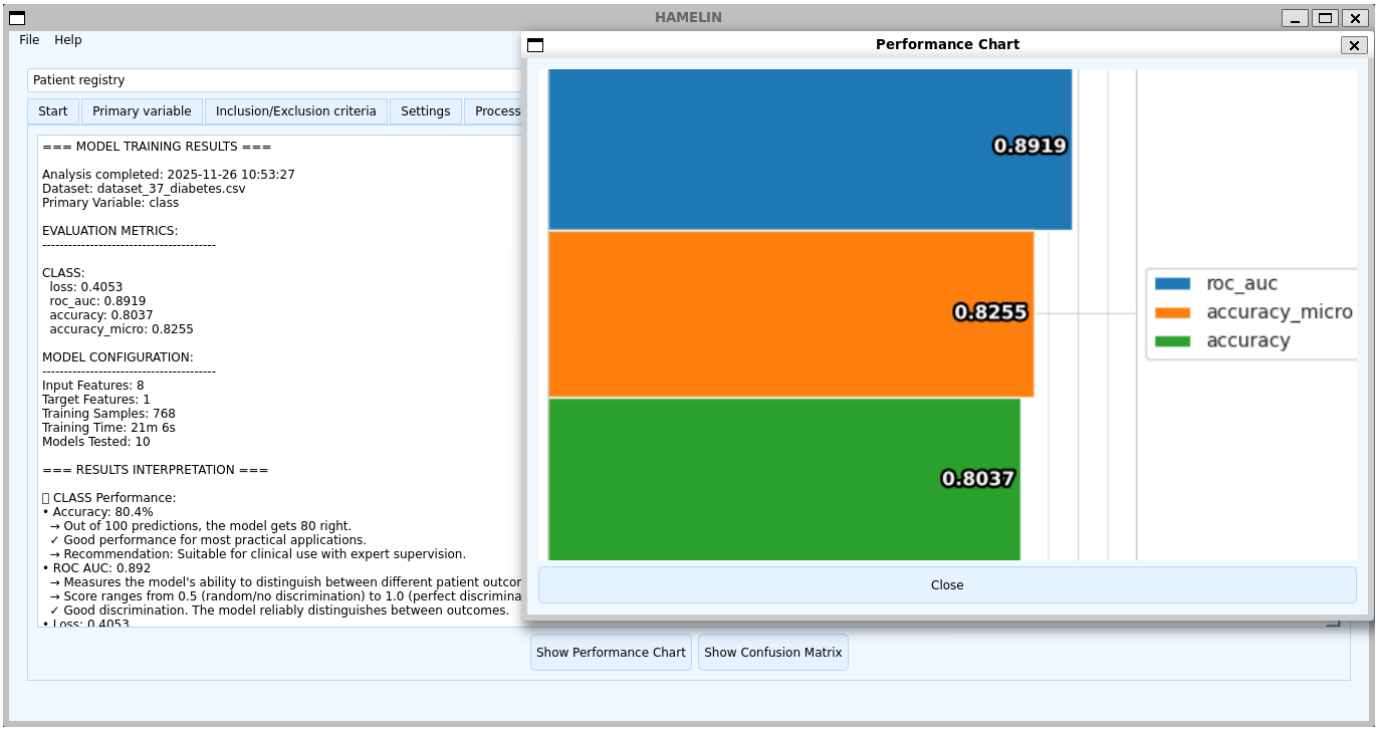



Fig. 5. Training outcomes and model performance.

## IV. Alignment with Human-Guided ML

To evaluate HAMELIN, we follow the HGML criteria [16], which provide a structured framework to assess how domain experts can guide data use, model development, and result interpretation in interactive ML systems. Applying these criteria highlights both the strengths and limitations of HAMELIN in supporting clinical researchers (Table I).

### A. Data Use

HAMELIN offers strong support for data-centric tasks. Clinicians can select variables (VS1) through the Primary Variable and Settings tabs, although without yet prioritizing variables based on clinical hypotheses. They can change temporal framing of variables (VS3) through the Criteria tab by defining protocol phases, though this feature is still under development. They can impute missing values (VS4) using Ludwig's comprehensive set of strategies, exposed in the Settings tab with clinical explanations. They can require inclusion or exclusion of variables (VR1) through the Criteria tab, which provides an intuitive rule builder for specifying complex inclusion/exclusion logic. They can select subpopulations of instances (IS2) and remove specific instances (IS3) through the same Criteria tab.

However, some data-centric tasks are not yet supported. Users cannot create new variables from existing ones (VS2) within the interface; this must be done through external preprocessing. They cannot augment data using new sources (VS5). They cannot select instances to label or annotate (IS1), as HAMELIN assumes that all data are already labeled. And they cannot define train/test split and cross-validation (IS4) through the Settings tab, choosing from k-fold, leave-one-out, or temporal split strategies.

### B. Model Development

Regarding model development, HAMELIN relies on the underlying AutoML backend (Ludwig) for model selection and hyperparameter tuning. Users have high-level control over preprocessing and training options but not detailed control over model choices. Ludwig automatically selects the model family (MS1) based on data characteristics and task type, but users cannot explicitly request a specific model type (e.g., "use only logistic regression"). Users cannot replace the model in a prior solution (MS2) or compare solutions with different model types (MC1) within the interface.

Users can request specific parameter values (PT1) through the Settings tab by defining custom hyperparameter search spaces, though this requires some familiarity with hyperparameters. Users can request specific data-preparation steps (SD1) and replace data-preparation steps (SD2) through the Settings tab by choosing missing data strategies.

TABLE I
MAPPING BETWEEN HGML TASKS [16] AND HAMELIN CLINICAL FUNCTIONALITIES.

| Category | HGML task | Where in HAMELIN | Implementation status |
|---|---|---|---|
| **Data Use** | | | |
| Variable Selection | [VS1] Variable selection (prioritise variables) | *Primary Variable*, *Settings* | Partially |
| | [VS2] Create new variables from existing ones | – (external preprocessing) | Not implemented |
| | [VS3] Change temporal / spatial framing of variables | *Criteria*; protocol phases | Partially |
| | [VS4] Impute missing values using other variables | *Settings* | Fully |
| | [VS5] Augment data using new sources | – | Not implemented |
| Variable Requirements | [VR1] Require inclusion / exclusion of variables | *Criteria* | Fully |
| Instance Selection | [IS1] Select instances to label / annotate | – | Not implemented |
| | [IS2] Select subpopulation of instances | *Criteria* | Fully |
| | [IS3] Remove specific instances | *Criteria* | Fully |
| | [IS4] Define train/test split and cross-validation | – | Not implemented |
| **Model Development** | | | |
| Model Selection | [MS1] Request a specific type of model | *AutoML Layer* (Ludwig) | Partially |
| | [MS2] Replace the model in a prior solution | – | Not implemented |
| Parameter Settings | [PT1] Request specific parameter values | *Settings*, Hyperparameter search (Ludwig) | Partially |
| Solution Design | [SD1] Request specific data-preparation steps | *Settings* | Fully |
| | [SD2] Replace a data-preparation step | *Settings* | Fully |
| **Model Interpretation** | | | |
| Model Assessment | [QC1] Request quantities of interest / metrics | *Results* | Fully |
| | [QC2] Request intermediate results | – | Not implemented |
| Model Comparison | [MC1] Compare solutions with different model types | – | Not implemented |
| | [MC2] Compare solutions on different subsets of instances | – | Not implemented |
| | [MC3] Compare solutions with same model but different data prep | – | Not implemented |
| | [MC4] Compare solutions with same model but different parameters | – | Not implemented |
| | [MC5] Compare two models with all other steps equal | – | Not implemented |
| Parameter Comparison | [PC1] Several solutions with same model and varying parameter ranges | – | Not implemented |

### C. Model Interpretation

In model interpretation, domain metrics are available. Users can request quantities of interest and metrics (QC1) through the Results tab, which presents accuracy, loss, area under the curve receiver operating characteristic (AUC_ROC), mean absolute error (MAE), root mean squared error (RMSE). However, users cannot request intermediate results (QC2), such as the performance of models evaluated during hyperparameter search.

Systematic comparison of alternative solutions is a significant limitation. Users cannot compare solutions with different model types (MC1), on different subsets of instances (MC2), with the same model but different data preparation (MC3), with the same model but different parameters (MC4), or two models with all other steps equal (MC5). They cannot generate several solutions with the same model and varying parameter ranges (PC1). These comparison tasks would require the system to maintain a history of all models trained and to provide interfaces for selecting and comparing them, which is not currently implemented.

## V. DISCUSSION

The development of HAMELIN demonstrates that protocol-aware AutoML systems can help bridge the gap between advanced ML techniques and their practical adoption in clinical research. By involving CRPs throughout the design process and grounding the interface in real clinical protocols, the system aligns more closely with established clinical research practices than generic AutoML tools.

This alignment is the result of several design decisions. First, the use of clinical terminology (e.g., “primary variable” instead of “target variable”, “criteria” instead of “filters”) improves accessibility and reduces cognitive friction for CRPs. Second, the tab-based workflow reflects the logical sequence of a clinical study, formulating the research question, preparing data, defining

the study population, configuring the analysis, executing the models, and interpreting the results, providing a familiar and intuitive structure. Third, explicit support for inclusion and exclusion criteria addresses a critical gap in existing AutoML systems and ensures that models are trained on valid populations. Finally, exposing Ludwig's missing data strategies with clinically grounded explanations enables users to make informed decisions about a complex preprocessing step.

Despite these strengths, the evaluation against HGML criteria reveals important limitations. While HAMELIN offers strong support for data-centric tasks, it provides limited functionality for model-centric and comparison tasks. Users cannot explicitly choose model families, compare alternative models, or systematically explore different modeling strategies. These limitations reflect a broader tension in AutoML system design between automation and user control. In its current form, HAMELIN prioritizes automation in model selection and hyperparameter tuning, delegating these decisions to Ludwig.

Future research should investigate how to introduce greater user control over modeling decisions without overwhelming CRPs with technical complexity. In addition, the current system lacks support for feature engineering within the interface. Clinical researchers often need to define derived variables (e.g., body mass index from weight and height, interaction terms, or nonlinear transformations) based on domain knowledge. At present, such transformations require external preprocessing, disrupting the workflow and requiring programming skills. Future versions of HAMELIN should therefore incorporate an interface for feature engineering that allows users to specify transformations using domain-meaningful concepts.

## VI. Conclusions

This work presents HAMELIN, a protocol-aware AutoML tool that enables clinical researchers without programming expertise to build predictive models while partially satisfying the criteria for HGML. By embedding protocol-aware constraints, the system ensures that predictive models respect temporal structures, endpoint hierarchies, and inclusion/exclusion criteria across patient registries, observational studies, and clinical trials. This design enhances clinical validity and coherence, distinguishing the approach from existing domain-agnostic AutoML platforms that treat clinical datasets as generic matrices.

The evaluation demonstrates strong support for data-centric operations, including variable selection, subpopulation definition, and inclusion/exclusion logic. However, model-centric tasks, such as explicit model family selection, hyperparameter specification, and systematic comparison of alternative solutions, remain partially constrained. Ongoing development aims to extend interactive control over these stages and incorporate advanced features such as feature engineering, provenance tracking, and multi-source data integration, moving the system toward full HGML compliance where domain expertise can guide all aspects of the modeling process.

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